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- Problem Statement ID – **SIH25063**
- Problem Statement Title- ***VOLT SENTINEL***
— *Smart Last-Mile Power Line Safety & Breakage Detection*
- Theme- **Disaster Management**
- PS Category- **Hardware**
- Team Name : **GridNova**

VOLT SENTINEL — Smart Last-Mile Power Line Safety & Breakage Detection



PROPOSED SOLUTION

- Deploy 20 sensor nodes along the 10 km EB line.
- Prototype: current sensor + voltage sensor + ESP32.
- Continuous telemetry to the EB office.
- Compare readings between adjacent nodes.
- Nodes 1–15 active, Nodes 16–20 offline → fault likely between Node 15–16.
- Voltage/current difference between adjacent nodes → leakage or tree contact location.
- Provides faster, section-level detection than manual inspection.

TECHNICAL APPROACH

Hardware: Current sensor, voltage sensor, ESP32, power supply, protected enclosure.

Communication: ESP32 Wi-Fi

Data flow: Sampling → Transmission → Node Comparison → Fault Alert

- ESP32 measures voltage and current.
- Server checks node status and signal continuity.
- Differential analysis identifies the affected span.
- Dashboard sends location, fault type, and time alert.

FEASIBILITY AND VIABILITY

- Uses low-cost, available sensors and ESP32 hardware.
- Modular design supports additional feeders.
- ESP32 Wi-Fi communication provides instant, low-latency node-to-server data transmission.

Risks and controls:

- Weather: sealed outdoor enclosure.
- Lightning: surge protection and proper earthing.
- Weak signal: Wi-Fi auto-reconnect routines.
- Sensor error: calibration and plausibility checks.
- False alarms: persistence timer and adjacent-node validation.

Pilot: Bench test → 3-node trial → 20-node field deployment.

IMPACT AND BENEFITS

Users: EB operators, maintenance teams, industrial customers, reliability planners.

- Reduce fault search from 10 km to one affected span.
- Reduce outage diagnosis and restoration time.
- Identify leakage and vegetation-contact points.
- Reduce maintenance crew exposure.
- Support prevention of electrical fires and unsafe conditions.
- Improve reliability reporting and customer updates.

Metrics: Node availability, alert time, false alarms, patrol distance, restoration time.

- **CEA Safety Regulations, 2023:**
Electrical safety, isolation, earthing, and overhead-line work 1 .
- **IEEE C37.114-2014:** AC transmission and distribution fault-location guidance 3.
- **IEEE 1366-2022:** Distribution reliability indices 4 .
- **ESP32 Wi-Fi Protocol:** High-speed real-time WebSocket and HTTP REST telemetry.
- **Prototype:** Current sensor + voltage sensor + ESP32.

References:

- [1] CEA Safety and Electric Supply Regulations
- [2] ESP32 Wi-Fi & TCP/IP Networking Documentation
- [3] IEEE C37.114-2014
- [4] IEEE 1366-2022